MIL-K-36486A
3 March 1969
SUPERSEDING
MIL-K-36486 (DM)
18 May 1966

MILITARY SPECIFICATION

KNIVES, PERIODONTAL, ORBAN

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers knives of Orban design.

*1.2 Classification. Knives shall be of the following styles, as specified (see 6.1).

Style 1 Orban, No. 19
Style 2 Orban, No. 20
Style 3 Orban, No. 21
Style 4 Orban, No. 22

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

* SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
WW-T-791	Tubing, Brass, Seamless.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

FSC 6520

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center.

24046	Knives, Periodontal, Orban.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The knives shall be in accordance with applicable requirements of GG-I-526, and as specified herein.

3.2 Material.

3.2.1 Operative ends. The operative ends shall be Class 4, either Type 440A, 440B or 440C stainless steel.

3.2.2 Handles. Handle No. 1 shall be Class 3, either Type 302, 303 or 304 stainless steel. Handle No. 2 shall be chromium or nickel plated brass and shall meet the requirements of WW-T-791, Grade 2, for composition only. Chromium plating with nickel intermediate shall be in accordance with GG-I-526 for brass. Nickel plating as a final coating shall be in accordance with the intermediate requirements of GG-I-526 for brass.

3.3 Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C52 to C60.

*3.4 Style, design and dimensions. The knives shall be Orban design knives which shall have a thin double-edged spear shaped cutting blade. The style, design and dimensions of the knives shall conform to Defense Personnel Support Center (DPSC) Drawing 24046 for the applicable style. Handles shall be furnished with handle and shank design of either Handle No. 1 or Handle No. 2. The knives shall be suitable for use in dissecting gingival tissue from anterior or posterior regions of the mouth during periodontal treatment.

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3.5 Construction. The knives shall be of two piece construction consisting of an operative end and handle. The handles shall be hollow. The operative ends shall be permanently secured to the handles. The finger grip portion of the handles shall be finely serrated or fine diamond knurled.

3.6 Finish.

3.6.1 Operative ends. The operative ends shall have a mirror or highly polished finish, blended to form a sharp cutting edge. The cutting edges shall be formed by the junction of the inner and outer surfaces of the blade.

3.6.2 Handles. The handles shall have a mirror or satin finish, except for the serrated or knurled portion. The serrated or knurled portion of the handles shall be free of barbs and shall provide adequate finger grip during use.

3.7 Corrosion resistance. The salt-spray test specified for brass composition in GG-I-526 shall not apply. The boil test shall be applicable to the entire instrument.

3.8 Identification marking. Each knife shall be marked with the word "Orban" followed by the respective instrument number, as applicable, for the style specified. Each instrument shall be marked on the front face. All marking shall be on the smooth portion of the handle.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I and III (see 4.2.1 and 4.3), and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. The unit of product for examination purposes shall be one knife.

Table I. Sampling for Examination

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-1	2.5
Minor defects	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, except that the acceptance number shall be zero for all sample sizes (see table II). Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table II. Sampling for Tests

Component	Characteristics	Rqmt.	Test Procedure	Inspection Level
Operative end	Hardness	3.3 and GG-I-526	GG-I-526	S-2
Operative end	Copper Sulfate	GG-I-526	GG-I-526	S-2
Entire knife	Boil test	3.7 and GG-I-526	GG-I-526	S-2
Entire knife	Composition	3.2.1 and 3.2.2	GG-I-526	*

*Sufficient amount from each lot of each metal used in the construction of the knives shall be selected for composition analysis.

4.3 Examination. The knives shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to the defects listed in table III.

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Table III. Classification of defects

Categories	Defects
Critical	None defined.
Major A:	
101	Style and design not as specified.
102	Instrument not constructed as specified.
103	Operative end finish not as specified.
104	Operative end not free from cracks or burrs.
Major B:	
121	Operative end dimensions not as specified.
Minor:	
201	Handle finish not as specified.
202	Instrument not permanently and legibly marked with either the name or trademark of the contractor.
203	Instrument identification not present or properly located.
204	Handle dimensions not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One knife, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each knife shall be wrapped in neutral tissue and packaged in a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors shall be acceptable. The carton shall be designed with suitable interior fittings to securely hold the instrument and prevent damage to the blade. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table III, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification. In addition, each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table II, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with the appendix of the applicable box specification.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style of knife (see 1.2)
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

6.3 Specification changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
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Review activities
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-K-36486A FSII

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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